Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Based Software for Brain Aneurysm Risk Stratification and Selective Neurovascular Imaging Triage			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio · Medical		Bio · Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	Intelligent Interpretation Systems	Medical Information & Systems	Remote and Home Health care Devices

Concept of Proposed Technology	The proposed technology is an **AI-based Software as a Medical Device (SaMD)** designed to support **brain (intracranial) aneurysm risk stratification and screening triage** using **non-imaging clinical variables** routinely captured in health screening data and real-world clinical records. The SaMD generates **patient-level risk scores** to identify individuals who may benefit from **selective confirmatory neurovascular imaging** such as CTA or MRA, enabling earlier identification and preventive management of intracranial aneurysm. This approach addresses the challenge that brain aneurysm is frequently asymptomatic until rupture, while population-wide neuroimaging is not feasible due to cost and resource constraints. The project focuses on generating **U.S.-relevant clinical evidence** through rigorous **external validation using independent U.S. datasets**, including diverse and multi-ethnic populations. Validation will rely on imaging-confirmed aneurysm status and/or longitudinal clinical outcomes, such as aneurysm diagnosis and aneurysmal subarachnoid hemorrhage events. Prespecified analyses will evaluate **generalizability, safety, effectiveness, and subgroup performance**. In parallel, the project will produce an **FDA-aligned, submission-ready evidence and documentation package**, including clinical evaluation, software lifecycle documentation, cybersecurity, risk management, and quality system readiness, to support U.S. regulatory authorization and commercialization.
Rationale for Joint R&D Project	**Unmet Clinical Need and Socioeconomic Burden** Intracranial aneurysm is often clinically silent until rupture, and aneurysmal subarachnoid hemorrhage is associated with high mortality, long-term neurological disability, and substantial downstream healthcare utilization. Earlier identification of individuals at higher risk presents an opportunity to mitigate severe outcomes and reduce socioeconomic burden. **Need for a Risk-Based Screening Triage Layer** With an estimated prevalence of approximately 2–3% in the general adult population, millions of individuals in the United States may harbor undiagnosed intracranial aneurysms. Definitive detection relies on neurovascular imaging modalities that are costly and operationally intensive, making universal screening infeasible. A **risk-based screening triage layer** is therefore necessary to prioritize individuals most likely to benefit from selective imaging and preventive intervention. **Necessity of Korea–U.S. Joint R&D** AI-based SaMD developed using Korean data requires **independent U.S. external validation** to demonstrate robustness, safety, and effectiveness in U.S. healthcare settings, which differ in population characteristics, care delivery, and data capture practices. A Korea–U.S. joint R&D framework enables compliant access to U.S. datasets under appropriate governance, supports rigorous subgroup analysis, and mitigates algorithmic bias. **Feasible Evidence Generation for Low-Prevalence Outcomes** Given the low prevalence of aneurysm rupture outcomes, real-world data–based external validation anchored by imaging-confirmed ground truth and/or longitudinal outcomes represents a feasible and scientifically appropriate approach. Joint R&D enables prespecified analysis plans, bias mitigation, and uncertainty reporting aligned with regulatory expectations. **Strategic Impact** The joint project supports generation of **regulatory-grade evidence** and accelerates U.S. regulatory readiness and commercialization, strengthening global competitiveness of Korean digital health technologies.

Key Words	1. Brain aneurysm risk stratification 2. AI-based Software as a Medical Device 3. Selective neurovascular imaging triage 4. U.S. multi-ethnic external validation 5. FDA-aligned clinical evidence

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (5)	**End of the Project**	Level (7)
R&D Objectives	• Establish U.S.-relevant clinical evidence for brain aneurysm risk stratification SaMD through external validation using independent U.S. datasets • Demonstrate generalizability, safety, and effectiveness across heterogeneous sites and prespecified demographic subgroups • Generate FDA-aligned clinical and technical documentation supporting regulatory submission readiness • Define and execute an appropriate U.S. regulatory pathway strategy • Develop evidence-linked plans for U.S. clinical workflow integration and post-market performance monitoring			
R&D Target Specifications	**Clinical Validation** • Independent U.S. datasets representative of real-world care settings • Imaging-confirmed aneurysm endpoints and/or longitudinal clinical outcomes • Demonstration of acceptable discrimination, calibration, and clinical utility • Prespecified subgroup performance analysis and bias mitigation documentation **Regulatory Readiness** • Completion of FDA-aligned clinical evaluation report • Software lifecycle, risk management, cybersecurity, and quality system documentation • Submission-ready regulatory dossier structure **Commercialization Readiness** • Defined U.S. clinical workflow integration points • Evidence-linked market entry strategy • Post-market performance monitoring plan			
R&D Plan	**Phase 1: Project Initiation and Regulatory Planning** • Finalize intended use, decision context, and risk classification • Define U.S. regulatory strategy and governance framework **Phase 2: U.S. Data Acquisition and Study Design** • Secure access to independent U.S. datasets • Develop prespecified validation and statistical analysis protocols **Phase 3: External Validation and Evidence Generation**			

	• Conduct U.S. external validation • Perform subgroup and equity analyses • Document results in regulatory-grade format **Phase 4: Regulatory Documentation and Submission Readiness** • Compile FDA-aligned clinical and technical evidence • Prepare submission-ready dossier and regulatory interactions **Phase 5: Commercialization and Post-Market Readiness** • Define U.S. clinical workflow integration • Develop market entry and lifecycle management plans

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Facilitate compliant access to U.S. clinical datasets • Lead study design, validation methodology, and statistical analysis • Execute or oversee external validation and subgroup evaluation • Interpret clinical results and contribute to regulatory-grade reporting
Expected R&R of Korean SME Partner	• Provide the AI SaMD and technical documentation • Manage algorithm version control and lifecycle documentation • Lead U.S. regulatory strategy and submission readiness • Integrate validation evidence into regulatory and commercialization plans

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Outcomes** • U.S.-validated clinical evidence demonstrating generalizability and safety of a Korea-developed brain aneurysm SaMD • Completion of an FDA-aligned, submission-ready evidence and documentation package • Operationalized readiness for U.S. clinical deployment and commercialization **Expected Impact** • Accelerated U.S. market entry for Korean AI-based medical technologies • More efficient and equitable selective neurovascular imaging utilization • Potential reduction in catastrophic aneurysm-related outcomes and downstream disability • Strengthened Korea–U.S. collaboration in digital health and regulatory science • Contribution to global best practices for AI SaMD validation and lifecycle management

6. References

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